

PD Data Interpretation Guide

Assumptions (Perfect Draft)

We interpret PD stats under a fixed setting: (1) card talent is fixed, (2) event context is fixed (park/modifiers), and (3) large volumes of games are available daily. Under these conditions, observed wOBA/SIERA variation is dominated by **random sampling noise** rather than changing talent or changing context.

Core statistical claim

For averages derived from repeated trials, typical uncertainty shrinks roughly with $1/\sqrt{n}$. So the jump from 200 AB → 600 AB is large; 3000 AB → 6000 AB is a smaller refinement.

Metric scope

wOBA is a linear-weights offensive rate statistic; it values outcomes (BB, 1B, 2B, 3B, HR) by run value rather than counting all hits equally.

SIERA is an ERA estimator intended to capture underlying pitching skill more than raw ERA.

What “tiers” mean in this guide

When we say **tier**, we mean a **cluster** of similarly-performing players separated by a **clear gap** from the next cluster. Tiers are more reliable than “exact rank” in smaller samples, and they stay useful even in large samples because they map directly to draft decisions (“take anyone in Tier 1” vs “avoid Tier 3 unless the build forces it”).

Example (pitching): suppose the best block of pitchers sits around **3.3–3.5 SIERA**, and the next major block sits around **4.2–4.5 SIERA**. That gap (≈ 0.7 – 1.2 SIERA) is a tier break: it’s large enough that you can draft confidently “above the line” and treat the lower block as a different quality class. By contrast, two pitchers at 3.38 vs 3.44 are usually “same tier” unless you have truly massive samples and a draft situation that needs that resolution.

How to use sample size during a draft

In PD drafts, you are making “good enough” decisions under time and pick constraints. Sample size changes what you should trust: **150–600 AB**: treat wOBA/SIERA as a **signal**. Use it to sort into tiers, identify outliers, and avoid obvious traps. Do not overfit close gaps. **1500–6000 AB**: treat wOBA/SIERA as a **measurement**. Smaller gaps are more likely to be real in PD context and can justify a pick when roster fit is otherwise similar. A useful mental model: “statistics don’t magically stabilize; they become more stable as sample grows.”

Breakpoints and drafting takeaways

Hitters (wOBA)	AB range	Draft takeaways	Practical draft move
Signal	150–300	Big gaps vs the PD baseline matter; small gaps are unreliable.	Use wOBA to avoid obvious weak bats; break ties with scarcity/defense/handedness.

Hitters (wOBA)	AB range	Draft takeaways	Practical draft move
Stronger signal	300–600	Tiering becomes reasonable; repeated above/below-baseline performance is informative.	When deciding between tiers, trust the direction; inside a tier, draft for team fit.
Decision-grade	600–1500	Most hitters will be close to their PD true level within a tight band.	Use meaningful gaps to justify reaching for a bat that completes your build (role/position).
Measurement	1500–6000+	Near-true-skill under PD conditions; smaller gaps are more often real.	If two picks fit similarly, take the wOBA edge when the gap is not microscopic.

Pitchers (SIERA)	IP range	Draft takeaways	Practical draft move
Signal	20–50	Good for spotting extremes (dominant arms vs landmines).	Avoid paying for tiny edges; use this mainly for eliminate/flag decisions.
Stronger signal	50–100	Tiering starters/relievers becomes usable.	Secure stability early if your build is volatile; otherwise hunt value.
Decision-grade	100–250	Reliable for most draft choices; clear gaps are likely real in PD.	Use meaningful SIERA gaps to choose “ace vs bat” when you need variance control.
Measurement	250–500+	Very stable PD true-skill sorting.	Justify early pitching only when the edge is real and your bat pool is deep; tiny gaps still imply same tier.

Interpreting OOTP/PD data vs real baseball

A lot of sabermetric “intuition” comes from MLB, where talent and context drift constantly (injuries, aging, role changes, changing opponents and parks). Perfect Draft changes the problem: **Stationary talent:** you can safely pool across days/events with the same PD settings. More data keeps improving the estimate instead of going stale. **Stationary context:** park/modifier consistency makes “PD baseline” comparisons much cleaner than MLB. **Different calibration:** wOBA weights and SIERA were developed on real baseball data. They still work as strong *relative* indicators in PD, but do not assume that an MLB threshold (e.g., “SIERA 3.80 is good”) maps perfectly to your PD baseline. **Practical implication:** interpret everything **relative to the PD baseline**, and draft using **tier gaps** (clusters separated by meaningful breaks), especially when your sample is only in the hundreds.

Sources

- FanGraphs Sabermetrics Library — Sample Size: <https://library.fangraphs.com/principles/sample-size/>
- FanGraphs Sabermetrics Library — The Beginner's Guide to Sample Size: <https://library.fangraphs.com/the-beginners-guide-to-sample-size/>
- FanGraphs Sabermetrics Library — wOBA: <https://library.fangraphs.com/offense/woba/>
- FanGraphs Sabermetrics Library — Linear Weights: <https://library.fangraphs.com/principles/linear-weights/>
- FanGraphs Sabermetrics Library — SIERA: <https://library.fangraphs.com/pitching/siera/>
- Baseball Prospectus — Introducing SIERA: <https://www.baseballprospectus.com/news/article/10027/introducing-siera-part-1/>
- OpenStax — Standard error ($\sigma/\sqrt{n}$) concept (Chapter Review): <https://openstax.org/books/introductory-business-statistics/pages/7-chapter-review>
- OOTP Manual — League Totals and Modifiers (context affecting statistical output): https://manuals.ootpdevelopments.com/index.php?man=ootp22&page;=ootp9-help_game_create_new_game_page_strategy